Understand the model first

Source slides 1-3 / No previous physics course is assumed.

The question behind every equation

A cart is already moving. We know how its velocity changes. How can we work out both its final velocity and its change in position?

Our assumptions

The motion is along one straight line. Acceleration stays constant. We choose the forward direction as positive. The cart keeps moving forward during the interval.

Meet each symbol before using it

Symbol	Quantity	Meaning	Unit
v_0	Initial velocity	How fast the cart moves at time zero.	$\frac{m}{s}$
a	Acceleration	How much velocity changes each second.	$\frac{m}{s^2}$
t	Elapsed time	Time since the motion starts.	s
$v(t)$	Velocity at time t	How fast the cart moves at the chosen time.	$\frac{m}{s}$
Δx	Displacement	Final position minus starting position.	m

The subscript 0 means **at the start**. The brackets in v(t) mean **the velocity evaluated at time t**; they do not mean multiplication. The Greek letter delta means **change**.

The logical route through the lesson

Acceleration tells us the velocity change. Add the starting velocity to get **final velocity**. The **velocity-time graph** then gives an area, which is **displacement**. These links solve the assignment on pages 6-7.

How acceleration changes velocity

Source slide 1 / First understand the definition, then rearrange it.

1. A rate means change divided by time

Acceleration is the change in velocity per unit of time. For constant acceleration, compare the start with a later time t.

$$a = \frac{v(t) - v_0}{t}$$

The numerator v(t) - v_0 is the velocity change. The denominator t is elapsed time; this rate equation requires t greater than zero. Here a is acceleration, v(t) is the later velocity and v_0 is the starting velocity. An acceleration of 3 m/s^2 adds 3 m/s of velocity every second.

2. Undo the division, then add the starting value

$$a\,t = v(t) - v_0$$

Multiply both sides by t. This preserves equality and turns a rate into the total velocity change over t seconds.

$$v(t) = v_0 + a\,t$$

Add v_0 to both sides. Final velocity equals the **starting velocity plus the velocity gained**.

3. Insert the values only after the relation is clear

$$v(4) = 2 + 3 \times 4 = 14\ \mathrm{m\,s^{-1}}$$

Read the answer in words

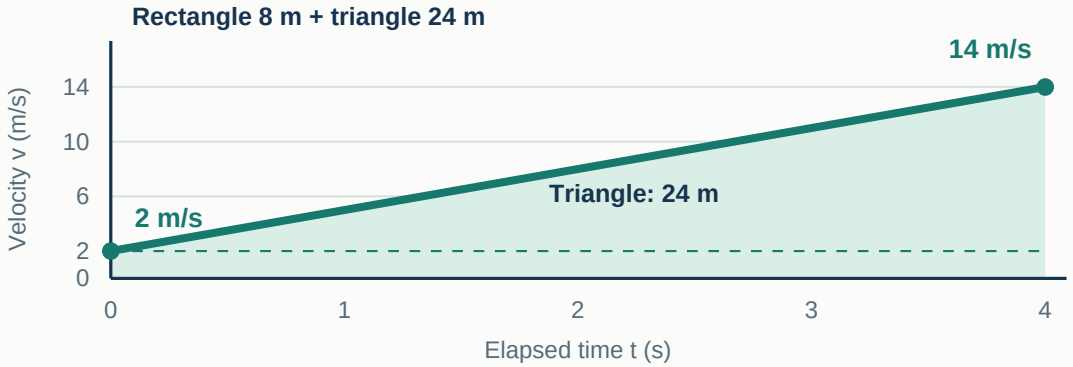
The cart starts at 2 m/s and gains 12 m/s over 4 s. Its velocity at the end is 14 m/s. The gained velocity is not the final velocity.

Turn velocity into displacement

Source slide 2 / The graph explains why the next formula works.

1. Why the area has the right meaning

At constant velocity, displacement equals velocity times time: the rectangle's area. For changing velocity, add thin time strips; the total area gives displacement.



2. A straight line makes the average easy

Constant acceleration makes velocity change linearly. The average velocity over this interval is halfway between the two endpoints. The new symbol v-bar denotes that average; v_0 and v(t) are the start and end velocities.

$$\bar{v} = \frac{v_0 + v(t)}{2} = \frac{2 + 14}{2} = 8\ \mathrm{m\,s^{-1}}$$

3. The average connects velocity to position

$$\Delta x = \bar{v}\, t = 8 \times 4 = 32\ \mathrm{m}$$

The whole area equals a rectangle at the average height: 8 m/s for 4 s. Delta x means displacement and t means elapsed time. Velocity multiplied by time gives metres.

Distance and displacement

Displacement is a signed position change. Distance is total path length. They have the same magnitude here because the cart never reverses direction.

Derive the displacement equation

Source slide 3 / Each line follows from a reason, rather than a rule to memorise.

1. Connect the two relations we already know

Page 3 gave displacement = average velocity multiplied by time. Page 2 gave final velocity = initial velocity + acceleration multiplied by time. Here delta x is displacement, v_0 is starting velocity, v(t) is final velocity, a is acceleration and t is elapsed time.

$$\Delta x = \frac{v_0 + v(t)}{2}\, t$$

2. Replace the final velocity with its expression

$$\Delta x = \frac{v_0 + (v_0 + a\,t)}{2}\, t$$

Substitution is valid because v(t) and v_0 + a t are equal. The brackets keep that whole replacement together.

3. Combine like terms, then distribute the time

$$\Delta x = \frac{2v_0 + a\,t}{2}\, t = v_0\, t + \tfrac{1}{2} a\, t^2$$

The two copies of v_0 combine to $2v_0$. Divide each numerator term by 2, then multiply each term by t. The acceleration term contains t multiplied by t, which is t^2.

Where the one-half comes from

On page 3's graph, v_0 t is the rectangle. The extra velocity grows from zero to a t, so its region is a triangle. A triangle has half the area of a rectangle with the same base and height.

Check the model and the units

Prerequisite bridge / Units carry meaning and expose mistakes before the answer is submitted.

1. Read each unit as a relationship

Metres measure displacement. Seconds measure elapsed time. Velocity is metres per second: distance for each second of motion. Acceleration is metres per second squared: velocity change for each second.

$$[v] = \frac{\text{m}}{\text{s}} \qquad [a] = \frac{\text{m}}{\text{s}^2} \qquad [t] = \text{s}$$

The square brackets here mean **the unit of the quantity**, not a new operation on the cart. This notation is only for checking dimensions.

2. Check each term of the equation separately

$$\frac{\text{m}}{\text{s}}\,\text{s} + \frac{\text{m}}{\text{s}^2}\,\text{s}^2 = \text{m} + \text{m}$$

Both terms describe a length, so adding them makes sense. The squared time cancels the squared seconds in acceleration's unit. One-half has no unit; it changes the amount, not the kind of quantity.

A mistake the units can reveal

If you forget to square t, the acceleration term has units of m/s. You cannot add that velocity to a displacement in metres. Checking units does not prove every arithmetic step, but it catches this structural error.

3. Check whether the assumptions still fit

The endpoint-average relation requires a straight velocity-time line, which follows from constant acceleration. Positive velocity throughout this interval means the cart moves forward without reversing. These facts explain both the formula and the match between distance and displacement.

A common trap: final velocity times time gives 56 m. That assumes the cart was already moving at 14 m/s throughout all four seconds. Use average velocity over the interval instead.

Read the assignment, build a plan

Integrated homework / The full question is included before any solution.

Original authored homework - exact question

A cart moves in a straight line with an initial velocity of 2 m/s. It has a constant acceleration of 3 m/s^2 in the same direction for 4 s. (a) Find its velocity after 4 s. (b) Find its displacement during these 4 s. (c) Verify the displacement using the area under a velocity-time graph.

1. Translate the sentence into the model

Given: v_0 = 2 m/s, a = 3 m/s^2, t = 4 s. Both velocity and acceleration point forward, so all three values are positive. **Find:** v(4), displacement delta x, and an independent graph check.

2. Choose the lesson link needed for each part

Part (a), velocity: use page 2, starting value + change.
Part (b), displacement: use page 4, rectangle + triangle.
Part (c), check: use page 3, area under the velocity-time line.

3. Solve part (a) one action at a time

$$v(4) = v_0 + a\,t$$

Write the relationship first. It matches part (a) because it predicts velocity from the initial velocity and its change.

$$a\,t = 3 \times 4 = 12\ \mathrm{m\,s^{-1}}$$

Multiply acceleration by elapsed time to find the velocity gained. Then add the initial 2 m/s.

$$v(4) = 2 + 12 = 14\ \mathrm{m\,s^{-1}}$$

Connection: the final 14 m/s supplies the graph's endpoint for part (c). Part (b) uses the displacement equation; the graph checks it.

Complete and check the assignment

Homework parts (b) and (c) / Connect calculation, graph and physical meaning.

4. Solve part (b): calculate the two contributions

$$\Delta x = v_0 t + \frac{1}{2} a t^2$$

Here delta x is displacement, v_0 initial velocity, a acceleration and t time. The terms separate starting motion from the extra motion caused by acceleration.

$$v_0 t = 2 \times 4 = 8 \text{ m}$$

$$\frac{1}{2} a t^2 = \frac{1}{2} \times 3 \times 4^2 = 24 \text{ m}$$

Square time first: 4^2 = 16. Then multiply by 3 and divide by 2. Finally add the two contributions: 8 + 24 = **32 m**.

5. Solve part (c): verify with the graph

Draw the line from (0 s, 2 m/s) to (4 s, 14 m/s) and split at 2 m/s. Here A means the area under the graph; its subscript names the shape.

$$A_{\text{rectangle}} = 4 \times 2 = 8 \text{ m}$$

$$A_{\text{triangle}} = \frac{1}{2} \times 4 \times (14 - 2) = 24 \text{ m}$$

The triangle's height is the **increase** in velocity: 14 - 2 = 12 m/s. The total graph area is 8 + 24 = 32 m, agreeing with part (b).

Interpret the complete answer

After 4 s, the cart moves at 14 m/s and is 32 m forward of its starting point. This is reasonable: its average velocity is 8 m/s, between 2 and 14; 8 m/s multiplied by 4 s gives the same 32 m.

Try it, explain it, check it

Retrieval practice / A new question tests transfer rather than copying the worked answer.

Try before reading the answer

A cart starts at 1 m/s and accelerates forward at 2 m/s^2 for 3 s. Find its final velocity and displacement. Explain why one-half appears in the displacement equation, then check the graph area.

Use the same chain with new numbers

Choose: v_0 = 1 m/s, a = 2 m/s^2, t = 3 s. **Predict:** final velocity should be greater than 1 m/s because acceleration is forward.

$$v(3) = 1 + 2 \times 3 = 7 \text{ m s}^{-1}$$

$$\Delta x = 1 \times 3 + \tfrac{1}{2} \times 2 \times 3^2 = 3 + 9 = 12 \text{ m}$$

The graph rectangle is 3 m; the triangle is 9 m. The triangle has half a rectangle's area because the extra velocity grows from zero to 6 m/s.

An independent check with average velocity v-bar

$$\bar{v} = \frac{1+7}{2} = 4 \text{ m s}^{-1} \quad \Rightarrow \quad \Delta x = 4 \times 3 = 12 \text{ m}$$

Explain the logic without looking at a formula

Acceleration gives the velocity change. Add the starting velocity to get the final value. With linear change, average the endpoints. Multiply that average by time to obtain displacement.

Model limits: varying acceleration can invalidate endpoint averaging. A reversal makes distance differ from signed displacement. Check both assumptions before reusing the formula.